TriAlign

An agentic tool for clinically meaningful comparison of trial eligibility criteria

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Clinical Protocols → Cleaned Criteria → Meaningful Comparison

Autocriteria, The Inspiration

AutoCriteria (2023)

- converts eligibility prose into structured criteria
- establishes scalable LLM-assisted preprocessing

Modernization

- ingests NCT records, text, and protocol PDFs
- extracts reviewable atomic structure and logic
- adds evidence, stable IDs, validation, correction, and recoverable exports

TriAlign (2026)

- retrieves comparable criteria across protocols
- distinguishes equivalence, overlap, direction, and contradiction
- combines exact tools, a learned encoder, contextual reasoning, uncertainty, and review

Practical use

A clinical team designing or operating a trial can select one current criterion, find how similar past trials defined it, inspect the operational differences, and decide whether its own rule is unusually narrow, broad, or inconsistent.

Extraction remains reviewable evidence rather than clinical ground truth; scanned PDFs require OCR outside the current application. Datta et al., AutoCriteria (2023); github.com/JonathanMa03/modern-autocrit

Why Improved Criteria Are Necessary

Eligibility criteria determine who can enter a trial, yet important differences are buried in long protocols and inconsistent prose.

- Study teams compare criteria for protocol design, feasibility, competitive intelligence, evidence synthesis, and amendment review.
- Keyword search finds shared words; ordinary embeddings often find shared topics. Neither guarantees shared eligibility meaning.
- Users need to know whether criteria are equivalent, narrower, overlapping, contradictory, related, or incomparable—with evidence and uncertainty.

Practical goal

Select a criterion, retrieve comparable criteria across protocols, inspect the differences, and trace the result back to its source.

Our Contributions

Pair	Embedding problem	Clinical relation
Age ≥ 18 vs. age < 18	nearly identical wording	disjoint populations
No cardiovascular disease vs. controlled hypertension permitted	same clinical topic	exception changes scope
No prior chemotherapy vs. none within 6 months	high lexical overlap	temporal rule differs
BP $> 145/95$ once vs. twice in 3 months	same measurement	repetition/timing differ

Three questions

Where do conventional embeddings fail? Can reviewed hard cases improve a dense encoder? Can an agent choose the right tools, quantify uncertainty, and escalate safely?

The Planned Protocol Corpus

50 oncology trials

Biomarkers, stage, histology, prior therapy, washout periods, performance status, organ function, and nested exceptions.

50 cardiovascular trials

Heart failure, atrial fibrillation, coronary/valvular disease, hypertension, stroke, thrombosis, PAD, and aneurysm.

- Extraction complete: 3,145 candidate criteria from 100 trials; atomic-criterion review and correction are in progress.
- Final validated count will be reported after review; extracted rows are not yet training examples.
- Chia may supplement terminology and examples, but does not supply gold pairwise labels.
- Every atom retains NCT ID, source version, parent statement, context, review, and validation.

Kury et al., Chia, Scientific Data 7, 281 (2020).

Verifiable Interpretable Relationships

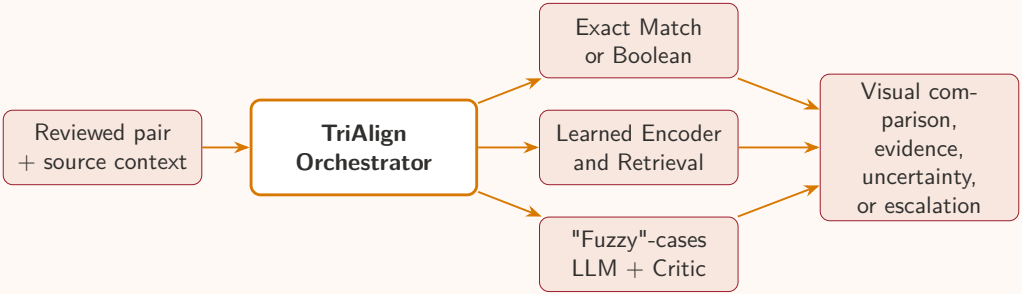
- **Equivalent** same operational rule
- **More restrictive** one permits a subset
- **Partially overlapping** shared but non-nested
- **Contradictory / disjoint** cannot both hold
- **Related, not comparable** shared topic, different operation
- **Unrelated** no meaningful relationship
- **Ambiguous** insufficient context

Why this matters

Similarity can rank candidates, but users see a clinical relationship, evidence, and calibrated uncertainty rather than a weighted mixture with unclear meaning.

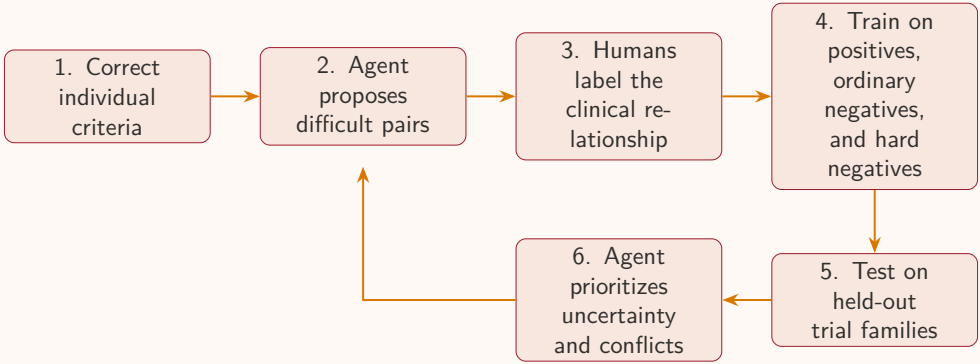
"More restrictive" means that one complete rule permits a subset of the other after inclusion/exclusion polarity and context are considered. Directional labels require a pairwise model; cosine distance alone is symmetric.

Where Do Agents Come In?



The orchestrator decides which route fits the comparison: exact computation for fully structured cases, encoder retrieval for learned clinical similarity, or contextual LLM review for fuzzy language. It may combine routes to check conflicts, then returns an evidence-linked visual comparison or sends the case to a human.

The Human Feedback Loop



Two different reviews

The current spreadsheet review validates each extracted criterion. The later pair review supplies labels such as equivalent, more restrictive, overlapping, disjoint, or ambiguous. Only accepted or adjudicated pair labels supervise the encoder.

Evaluation of TriAlign

Question	Test
How good are retrieved clinically comparable criteria?	Compare TF–IDF, frozen general/biomedical encoders, and our fine-tuned encoder using precision/recall@k, reciprocal rank, and anchor ordering.
Are important differences preserved?	Report errors by negation, thresholds, units, timing, repetition, exceptions, and logic; evaluate the separate relationship classifier with macro-F1 and order consistency.
Does the agent know when to escalate?	Measure calibration, risk–coverage, conflict detection, inappropriate acceptance, reviewer corrections, time, latency, and cost.
Did each component help?	Remove hard negatives, human corrections, critic checks, or targeted sampling and compare paired results across multiple seeds.

Trials are split by family before pairing; held-out labels are unavailable to agent retrieval. Natural and constructed cases are reported separately, with uncertainty preserving trial-family dependence.

Benchmark Comparisons

System	Implementation	Comparison role
Lexical baseline	TF–IDF + cosine	Word-overlap reference.
General encoder	`all-mpnet-base-v2`	Frozen semantic-retrieval baseline.
Biomedical encoder	`BioLORD-2023`	Frozen biomedical-representation baseline.
TrialGPT-style agent	Criterion-level LLM reasoning adapted from patient–trial matching	Tests an LLM-only evidence-and-judgment workflow on compatible criterion-pair cases.
EligMeta-style agent	LLM parsing plus deterministic eligibility-alignment logic	Tests recent eligibility-aware agent design on compatible alignment cases.
TriAlign	Fine-tuned MPNet + directional head + routed tools	Learns reviewed hard cases and chooses deterministic, embedding, or fuzzy comparison.

Fair comparison

All runnable systems receive the same reviewed pairs and evidence. We compare labels, ranking, evidence support, abstention, latency, and cost only where outputs align; TrialGPT and EligMeta remain task-adjacent rather than identical benchmarks.

Jin et al., TrialGPT, Nature Communications (2024); Zhao et al., EligMeta, arXiv:2604.02678 (2026); Hugging Face model cards for MPNet and BioLORD.

Five milestones from current review to delivery

	Stage	Timing	Exit artifact
1	**Freeze inputs**	Oct. 20–Nov. 2	Reviewed, blocker-free criterion corpus with versioned trial-family splits.
2	**Build benchmark**	Nov. 3–9	Adjudicated natural and challenge pairs plus frozen baseline results.
3	**Improve distance**	Nov. 10–16	Fine-tuned MPNet, directional relationship head, and validation-selected checkpoint.
4	**Complete agent**	Nov. 17–30	Routed exact/encoder/LLM tools, critic checks, calibration, abstention, and review workflow.
5	**Validate + deliver**	Final period	Held-out tests, agent comparisons, ablations, intervals, UI, documentation, and demonstration.

Gate between stages

A stage advances only after its versioned artifacts and acceptance checks pass. Incomplete review or split leakage blocks training instead of being silently ignored.

A Quick UI "Sketch"

User request

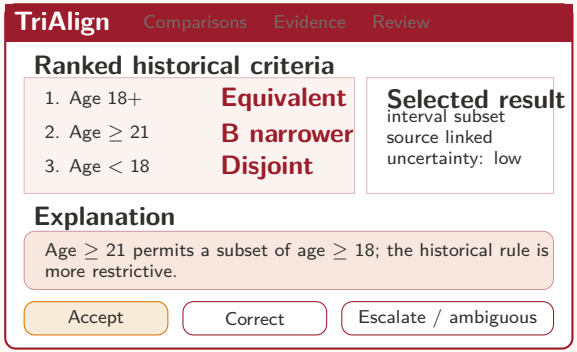
Current trial: NCT-new

Criterion: Age $\geq$ 18 years

Ask: Find comparable historical rules and explain the differences.

$\Downarrow$

Agent routes tools and retrieves evidence



The dashboard separates retrieval score from clinical relationship, exposes source evidence and agent checks, and preserves every human correction in the audit trail.